

# Practical Recommendations for Constructing Machine Learning Models for Biomedical Signal Processing

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**Abstract** — Machine learning (ML) algorithms can be very effective in biomedical signal processing, but their utility depends on following good methods of training and testing. Based on our experience, we present practical guidance to help researchers achieve consistent model reliability.

**Clinical Relevance** — Making reliable predictions in clinical settings will enhance patient care and safety. The proposed practical recommendations enable developers to increase the credibility and reliability when using ML models.

## INTRODUCTION

- ML can process biomedical signals in higher dimensionality and greater capacity [1], but model development requires specialized expertise.
- Practical guidelines on how to build them are lacking [1] and gradually emerging [2,3].
- We present recommendations on how to build ML models tailored for biomedical signal processing applications to improve credibility, trustworthiness of prediction models.

## METHODS

- $D=\{x_i, y_i\}$  is a dataset of  $n$  data points of  $m$  features
- $X = x_i$  are features and  $Y=y_i$  are outcomes for  $i=1,2,\dots,n$ .
- A model  $f: X \rightarrow Y$  may be a classification or regression.
- ML algorithms assume  $(x_i, y_i)$  are independently drawn from the same joint probability distribution  $P(X, Y)$  where:

$$P(X, Y) = P(Y|X) * P(X)$$

The above is rewritten as i.i.d assumptions [5]:

- $(x_i, y_i)$  is independent of  $(x_j, y_j)$  for all  $i, j=1,2,\dots,n$  and  $i \neq j$
- $(x_i, y_i)$  is identically distributed as  $P(X, Y)$  for all  $i, j=1,2,\dots,n$

- In biomedical applications, the above are often violated
- Violating 1** can introduce **bias** and **data leakage** [4]
- Violating 2** may increase **prediction errors** [5].
- When building an ML pipeline, **we recommend the experiment design of Fig. 1** supported by [3,6].

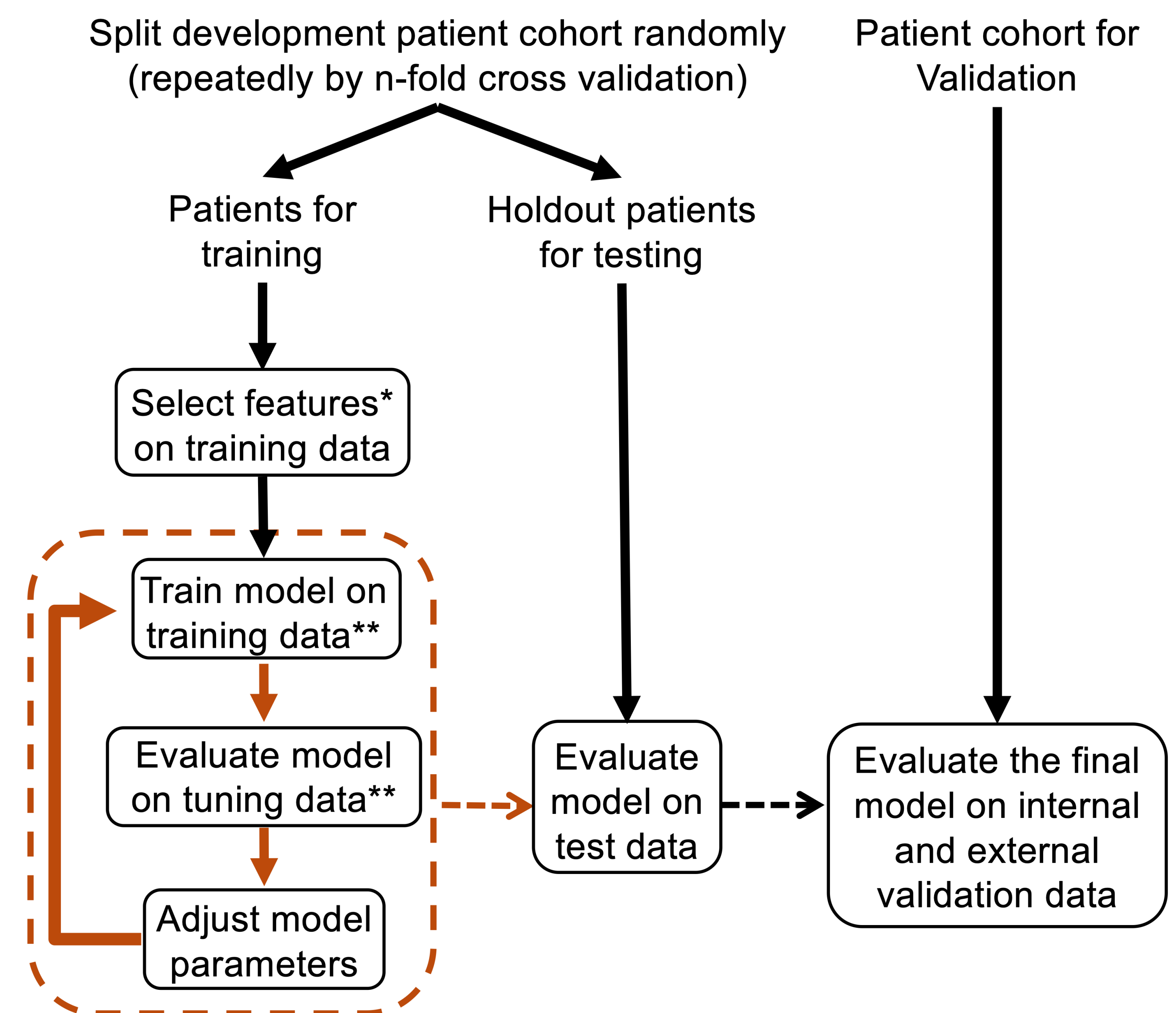
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\* Ideally, feature selection should be performed on a separate, independent patient cohort  
\*\* Prior to model training, reserve a small fraction of training data for model tuning

Figure 1- Proposed recommendations for model development.

## DISCUSSION & CONCLUSION

- Data leakage** occurs when information external to training data is used for building the model (violates assumption 1).
- ML models should only be **tested on unseen data** from patients, i.e., not included in training.
- Feature selection, model tuning, adjustment and fitting should only be carried out only on training data [4].
- The final model should be **validated on unseen subjects** drawn from the same institution (internal) or from other centers (external) to ensure compliance with assumption (Institutional variations may limit model generalizability).

**The above recommendations will establish reliable performance and trustworthy evaluation.**

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